

UNIT IV – MEMORY ORGANIZATION

Memory: Basic concept and hierarchy, semiconductor RAM memories, 2D & 2 1/2D memory organization. ROM memories. Cache memories: concept and design issues & performance, address mapping and replacement Auxiliary memories: magnetic disk, magnetic tape and optical disks Virtual memory: concept implementation.

◇ 1. Introduction to Memory

Memory is a **storage space** in the computer where data and instructions are kept for processing.

It plays a **vital role** in computer performance — faster memory → faster system.

Memory is Needed Because:

- CPU needs data & instructions to execute programs.
- Memory holds these temporarily or permanently.
- Acts as the **bridge between CPU and storage devices**.

◇ 2. Memory Hierarchy

The **Memory Hierarchy** represents memory arranged in order of **speed, cost, and capacity**.

Hierarchy Levels:

Level	Type	Speed	Cost	Capacity	Example
1	Registers	Fastest	Highest	Very small	CPU Registers
2	Cache	Very fast	High	Small	L1, L2, L3 Cache
3	Main Memory (RAM)	Medium	Moderate	Medium	DDR4, DDR5
4	Secondary Storage	Slow	Low	Large	Hard Disk, SSD
5	Tertiary Storage	Very Slow	Very Low	Very Large	Magnetic Tape, Optical Disk

Concept:

As we go **down the hierarchy**, speed decreases, capacity increases, and cost per bit decreases.

◇ 3. Semiconductor RAM Memories

RAM = **Random Access Memory**

→ You can access any memory location **directly and equally fast**.

Types of RAM:

Type	Full Form	Characteristics	Example
SRAM	Static RAM	Uses flip-flops; faster, expensive, no refresh needed	Used in Cache
DRAM	Dynamic RAM	Uses capacitors; slower, cheaper, needs refresh	Used in Main Memory

Comparison:

Feature	SRAM	DRAM
Storage Element	Flip-flop	Capacitor
Speed	Very Fast	Slow
Cost	Expensive	Cheap
Refresh Required	No	Yes
Density	Low	High
Use	Cache memory	Main memory

◇ 4. Memory Organization

Memory cells are arranged in a **matrix (rows × columns)**.

➤ 2D Memory Organization:

Each cell stores **1 bit** (0 or 1).

Organized in **rows (address lines)** and **columns (data lines)**.

Example: 1K × 8 memory = 1024 rows and 8 columns → 8-bit data word.

Diagram (conceptual):

Address Decoder → Row → [Memory Cell Array] → Sense/Write Circuits → Data Output

➤ 2.5D Memory Organization:

Improved 2D with **extra interconnections and buffer layers** (used in modern high-density memories).

Enhances **speed and data bandwidth**.

Used in **3D-stacked DRAMs** (like HBM – High Bandwidth Memory).

◇ 5. ROM Memories

ROM = **Read Only Memory**

→ Data is **permanently stored** and cannot be changed easily.

Types of ROM:

Type	Description	Reprogrammable?	Example Use
ROM	Programmed during manufacturing	✗ No	BIOS
PROM	User-programmable once	✗ No	Firmware
EPROM	Can be erased using UV light	✓ Yes	Microcontroller firmware
EEPROM	Can be erased electrically	✓ Yes	BIOS updates
Flash Memory	Faster, block-level erase	✓ Yes	Pendrive, SSD

◇ 6. Cache Memory

Cache = **small, high-speed memory** placed between **CPU and main memory (RAM)** to reduce access time.

Concept:

CPU first checks data in **cache**.

If found → **Cache Hit**

If not → **Cache Miss**, data fetched from main memory.

Example:

If CPU wants to access variable X:

Check cache — if X is there → use it (hit).

If not → fetch from RAM and store it in cache for next time (miss).

Cache Design Issues:

Aspect	Description
Cache Size	Larger cache → more hits but higher cost
Block Size	Size of data transferred from main memory to cache
Mapping	Determines where data from main memory is placed in cache
Replacement Policy	Decides which block to replace when cache is full
Write Policy	How to handle data modification (write-through/write-back)

Cache Address Mapping Techniques:

Type	Description	Advantage	Example
------	-------------	-----------	---------

Direct Mapping	Each block of main memory maps to only one cache line	Simple, fast	(Block address) mod (No. of cache lines)
Associative Mapping	Any block can go to any cache line	Flexible, higher hit rate	Expensive hardware
Set-Associative Mapping	Memory divided into sets; block maps to one set	Balance of speed & cost	Common in CPUs (e.g., 4-way set associative)

Cache Replacement Policies:

Policy	Description	Example
FIFO (First-In-First-Out)	Replace oldest block	Simple but less efficient
LRU (Least Recently Used)	Replace block least recently used	Most used method
Random	Replace any random block	Simple hardware

◇ 7. Auxiliary Memory

Auxiliary (Secondary) memory = **external non-volatile storage** used for **permanent data storage**.

Type	Working Principle	Example	Characteristics
Magnetic Disk	Data stored magnetically on rotating platters	Hard Disk Drive	Random access, high capacity
Magnetic Tape	Plastic tape coated with magnetic material	Backup tapes	Sequential access, cheap
Optical Disk	Data stored using laser light	CD, DVD, Blu-ray	Durable, portable

Example:

Hard disk → fast, used for OS & applications

Tape → used for backups

Optical disks → used for media distribution

◇ 8. Virtual Memory

Virtual Memory allows the system to use **part of secondary storage (disk)** as if it were **main memory**.

It helps run **large programs** even when physical RAM is small.

Working:

Program divided into **pages**.

Main memory divided into **frames**.

When a page needed → loaded into RAM.

If memory full → **Page Replacement** occurs.

Common Algorithms:

Algorithm	Description
FIFO	Remove oldest page
LRU	Remove least recently used page
Optimal	Remove page that will not be used soon

Real-life Example:

If your laptop has **8GB RAM** and you open a program needing **12GB**,

→ OS uses **virtual memory (pagefile/swap space)** on the hard disk to compensate.

◇ 9. Summary of Memory Types

Memory Type	Speed	Volatility	Use	Example
Register	Fastest	Volatile	Temporary CPU data	PC, MAR
Cache	Very Fast	Volatile	Stores recent instructions	L1/L2 Cache
Main Memory (RAM)	Medium	Volatile	Running programs	DRAM
ROM	Fast	Non-volatile	Permanent instructions	BIOS
Secondary	Slow	Non-volatile	Long-term storage	HDD
Virtual Memory	Slow	Virtual (disk-based)	Extend RAM	Pagefile

◇ 10. Key Differences (Quick Recall)

Feature	RAM	ROM
Data Access	Read/Write	Read-only
Volatile	Yes	No
Speed	High	Medium
Example	DRAM, SRAM	PROM, EPROM

Feature	Cache	Main Memory
Location	Between CPU and RAM	External to CPU
Speed	Faster	Slower
Size	Small	Large

1. Primary Memory vs Secondary Memory

Feature	Primary Memory	Secondary Memory
Definition	Directly accessible by CPU (RAM, ROM).	Used for permanent data storage (HDD, SSD, etc.).
Speed	Very fast.	Slower.
Volatility	Mostly volatile (RAM).	Non-volatile.
Cost per Bit	High.	Low.
Capacity	Limited (few GBs).	Very large (TBs).
Use	Stores data currently being processed.	Stores data and programs permanently.

? 2. SRAM vs DRAM

Feature	SRAM (Static RAM)	DRAM (Dynamic RAM)
Storage Element	Flip-flop.	Capacitor.
Speed	Faster.	Slower.
Cost	Expensive.	Cheap.
Refresh Needed	No.	Yes (periodic refresh).
Density	Low.	High.
Use	Cache memory.	Main memory.

3. ROM vs RAM

Feature	ROM (Read Only Memory)	RAM (Random Access Memory)
Data Access	Read-only.	Read/Write.
Volatility	Non-volatile.	Volatile.
Usage	Permanent program storage (firmware).	Temporary data while processing.
Data Stored During	Manufacturing/programming.	Program execution.
Example	BIOS, EEPROM.	DRAM, SRAM.

4. Cache Memory vs Main Memory

Feature	Cache Memory	Main Memory (RAM)
Location	Between CPU and RAM.	Between cache and secondary storage.
Speed	Faster.	Slower.
Size	Small (KB–MB).	Larger (GBs).
Cost	High.	Moderate.
Purpose	Reduce memory access time.	Store programs/data currently running.
Technology	SRAM.	DRAM.

5. Direct Mapping vs Associative Mapping vs Set-Associative Mapping

Feature	Direct Mapping	Associative Mapping	Set-Associative Mapping
Mapping Rule	Each block maps to a fixed cache line.	Any block can go to any cache line.	Block maps to any line within a specific set.
Flexibility	Least flexible.	Most flexible.	Medium flexibility.
Speed	Fast.	Slow (due to search).	Moderate.
Cost	Low.	High.	Moderate.
Hit Ratio	Lowest.	Highest.	Medium to High.

6. Virtual Memory vs Cache Memory

Feature	Virtual Memory	Cache Memory
Purpose	Extend main memory using disk space.	Speed up CPU-memory access.
Location	Part of secondary storage (HDD/SSD).	Between CPU and main memory.
Managed By	Operating System.	Hardware (CPU).
Data Type	Stores parts of programs not in RAM.	Stores frequently used instructions/data.
Speed	Slower (uses disk).	Very fast (SRAM).

7. Sequential Access vs Random Access Memory

Feature	Sequential Access	Random Access
---------	-------------------	---------------

Access Method	Data accessed in sequence.	Any location can be accessed directly.
Speed	Slow.	Fast.
Example	Magnetic tape.	RAM.
Use	Backup storage.	Working memory.

8. Magnetic Disk vs Optical Disk

Feature	Magnetic Disk	Optical Disk
Data Storage Method	Magnetic fields.	Laser light.
Access Type	Random access.	Sequential/random.
Durability	Less durable (mechanical).	More durable.
Rewrite Capability	Easy (HDD).	Limited (CD-RW, DVD-RW).
Example	HDD.	CD, DVD, Blu-ray.

9. 2D Memory Organization vs 2.5D Memory Organization

Feature	2D Organization	2.5D Organization
Structure	Simple planar (rows × columns).	Interconnected layers with interposer or buffer.
Speed	Moderate.	Higher due to shorter interconnects.
Density	Normal.	High (more data per area).
Used In	Basic RAM chips.	Modern high-bandwidth DRAM (HBM).

10. EEPROM vs Flash Memory

Feature	EEPROM	Flash Memory
Erase Method	Byte-by-byte erase.	Block-level erase.
Speed	Slower.	Faster.
Cost	Higher.	Lower.
Endurance	Higher (more write cycles).	Lower.
Use	BIOS, embedded systems.	Pendrives, SSDs.

Notes by -[GKM](#)